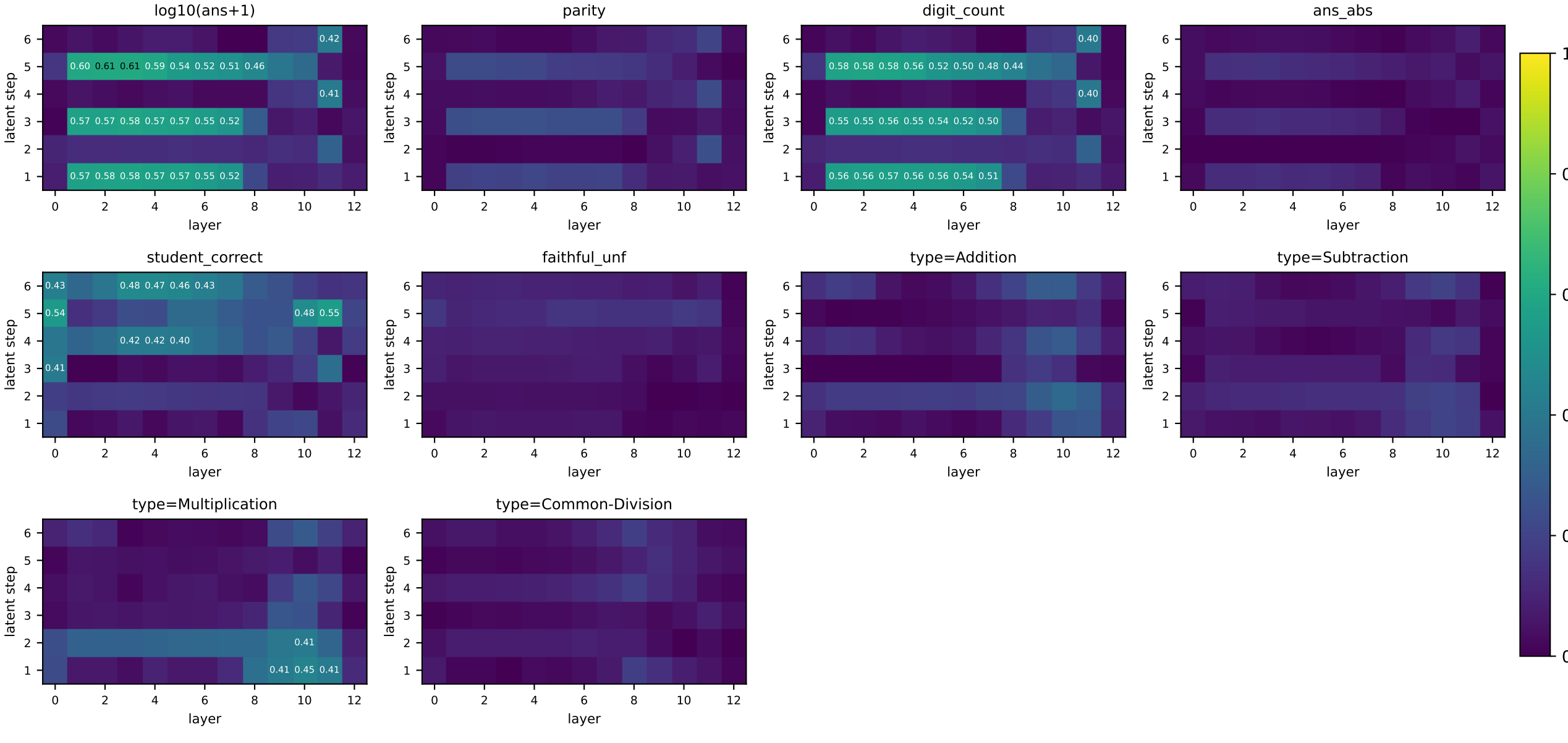
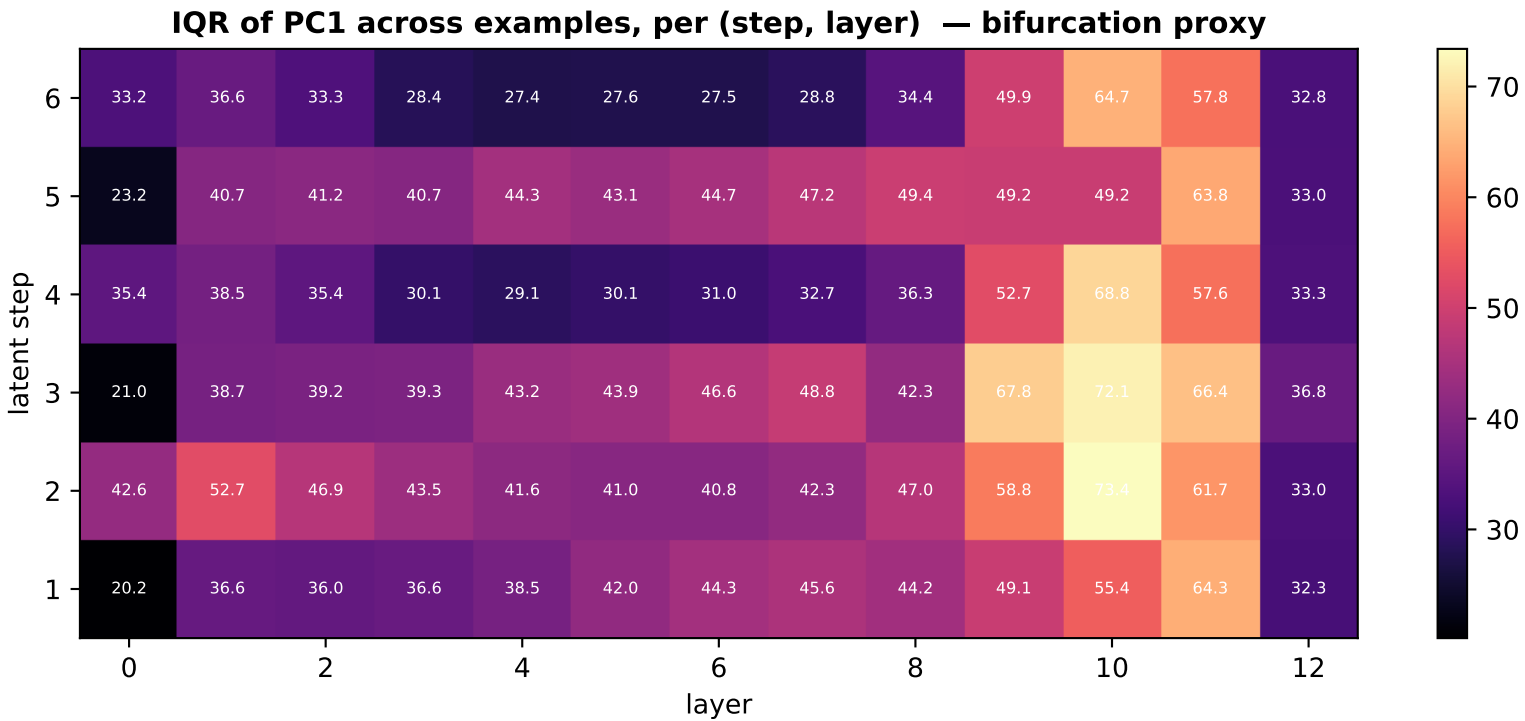
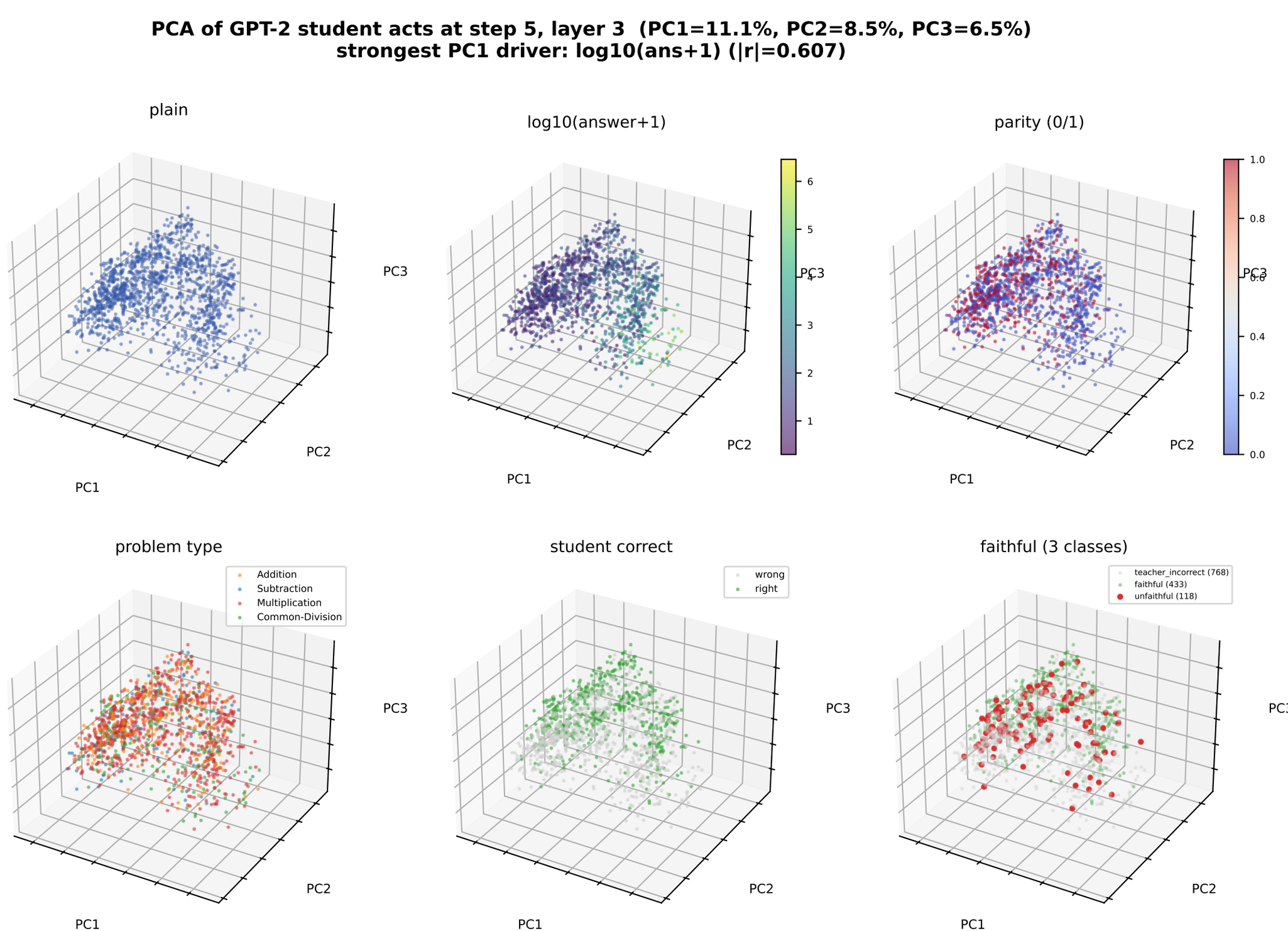
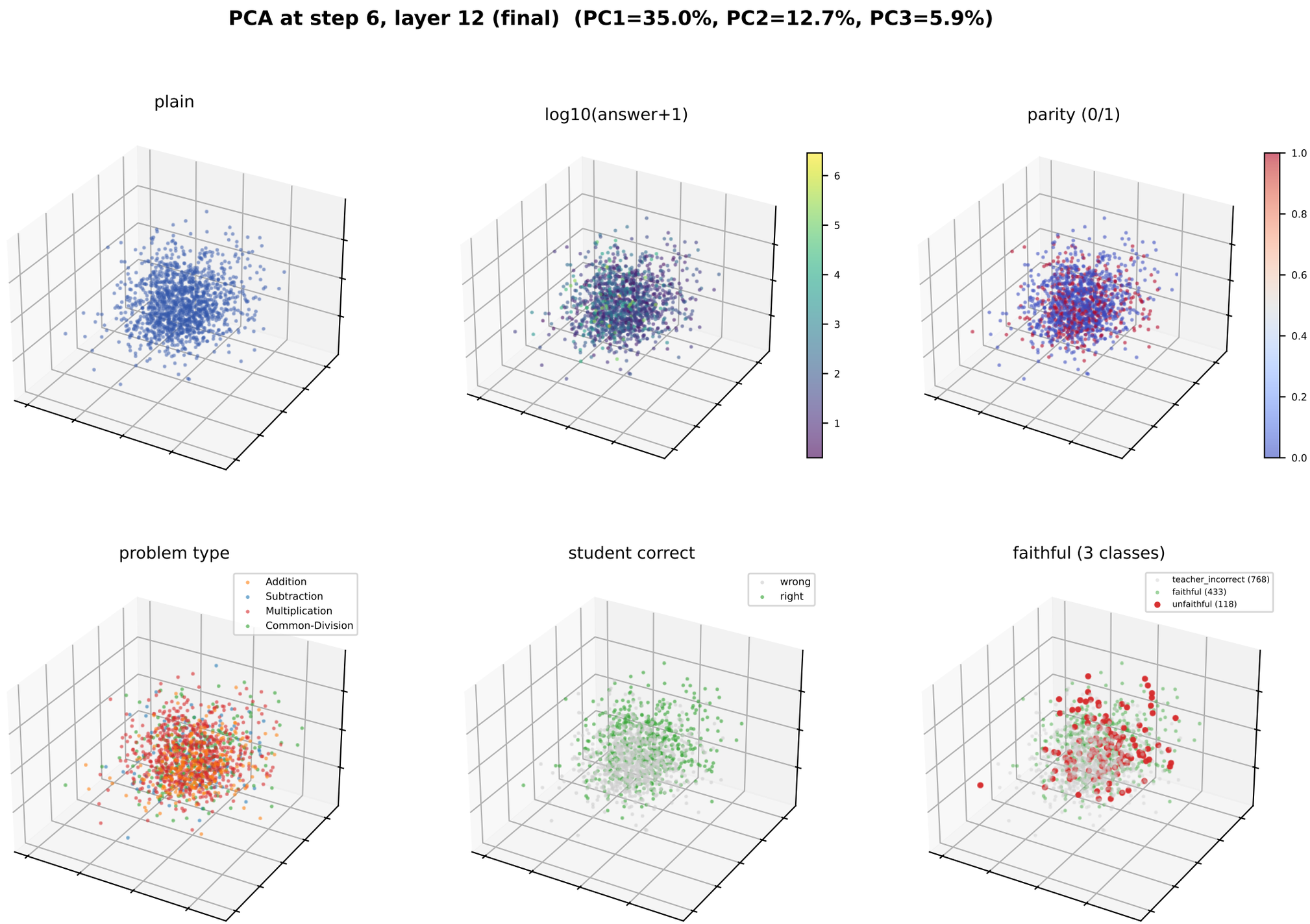
|Pearson r| between PC1 and candidate features, per (step, layer)




IQR of PC1 across examples, per (step, layer) — bifurcation proxy
latent step
layer
6: 33.2 36.6 33.3 28.4 27.4 27.6 27.5 28.8 34.4 49.9 64.7 57.8 32.8
5: 23.2 40.7 41.2 40.7 44.3 43.1 44.7 47.2 49.4 49.2 49.2 63.8 33.0
4: 35.4 38.5 35.4 30.1 29.1 30.1 31.0 32.7 36.3 52.7 68.8 57.6 33.3
3: 21.0 38.7 39.2 39.3 43.2 43.9 46.6 48.8 42.3 67.8 72.1 66.4 36.8
2: 42.6 52.7 46.9 43.5 41.6 41.0 40.8 42.3 47.0 58.8 73.4 61.7 33.0
1: 20.2 36.6 36.0 36.6 38.5 42.0 44.3 45.6 44.2 49.1 55.4 64.3 32.3
0 2 4 6 8 10 12
70 60 50 40 30


PCA of GPT-2 student acts at step 5, layer 3 (PC1=11.1%, PC2=8.5%, PC3=6.5%)
strongest PC1 driver: log10(ans+1) (|r|=0.607)
plain
log10(answer+1)
parity (0/1)
problem type
student correct
faithful (3 classes)
Addition
Subtraction
Multiplication
Common-Division
wrong
right
teacher_incorrect (768)
faithful (433)
unfaithful (118)
PC1
PC2
PC3


PCA at step 6, layer 12 (final) (PC1=35.0%, PC2=12.7%, PC3=5.9%)
plain
log10(answer+1)
parity (0/1)
problem type
Addition
Subtraction
Multiplication
Common-Division
student correct
wrong
right
faithful (3 classes)
teacher_incorrect (768)
faithful (433)
unfaithful (118)